

Design Review: Database Load Reduction

Author: Alex Chen | Date: Design Review Meeting

1. Proposal

We propose introducing a Redis cache-aside layer between the API service and the PostgreSQL primary database. Reads first check Redis; on a miss, the API reads from PostgreSQL and populates the cache. This is expected to reduce database load substantially and cut p50 read latency.

2. Architecture

API -> Redis Cache -> PostgreSQL

3. Trade-offs

Introducing a cache adds operational complexity and a small risk of serving stale data if the TTL is misconfigured. The cache TTL is set to 5 minutes as a starting point. We accept this risk given the significant latency improvement it provides. Cache invalidation timing was flagged as the primary open risk during the review.

4. Next Steps

Roll out to staging first, monitor cache hit rate and stale-read incidents closely for two weeks before promoting to production.